

Notes: Fulghieri, Strobl, and Xia (RFS, 2014)
The Economics of Solicited and Unsolicited Credit Ratings

Contents

1	Summary	3
2	Big picture	3
3	Incremental contribution of the paper	4
4	Model environment	4
4.1	Agents	4
4.2	Project types	4
4.3	CRA types and reputation	5
4.4	Two rating systems	5
5	Solicited-only rating system	5
5.1	Why solicited-only ratings create inflation	5
5.2	Equilibrium result	5
6	Rating system with unsolicited ratings	6
6.1	Why an unfavorable unsolicited rating is a credible threat	6
6.2	Equilibrium result	6
6.3	Selection rather than punishment bias	6
7	Fees, rating inflation, and welfare	6
7.1	Fee effect	6
7.2	Rating-inflation effect	7
7.3	Welfare effect	7
7.4	Comparative statics	7
8	Notes-created summary tables	8
8.1	Two systems	8
8.2	Key objects in the model	8
8.3	Main results	9
8.4	Mechanism map	9
9	Figures and formal results	9

9.1	Title, abstract, and motivating puzzle	9
9.2	Decision trees	10
9.3	Reputation and unrated firms in the solicited-only system	11
9.4	Success, failure, and CRA profit	11
9.5	Proposition 1	12
9.6	Unsolicited-rating objective and Proposition 2	13
9.7	Corollary 1 and fee/rating-inflation effects	13
9.8	Welfare and comparative statics	14
9.9	Conclusion	15
10	How to remember the paper	16
11	Conclusion	16
11.1	Core conclusion	16
11.2	Why the result matters	17
11.3	Takeaway	17

1 Summary

Fulghieri, Strobl, and Xia study why credit rating agencies issue **unsolicited** ratings and why these ratings are often lower than solicited ratings. The paper builds a dynamic rational-expectations model of the credit rating process with three core elements: rating agencies can misreport issuer quality, they can issue unsolicited ratings, and their current rating choices affect their future reputation.

The central mechanism is a dynamic trade-off for an opportunistic credit rating agency (CRA). If the CRA inflates a rating today, it can earn a high current fee from the issuer. But if a bad project later fails after receiving a high rating, investors update negatively about the CRA's type, lowering the credibility and fee-generating value of its future ratings. Reputation therefore disciplines rating inflation, but imperfectly, because investors cannot perfectly distinguish bad luck from bad rating policy.

Unsolicited ratings change this trade-off. The CRA can threaten issuers that refuse to buy a solicited rating with an unfavorable unsolicited rating. This threat raises the issuer's willingness to pay for a favorable solicited rating. At the same time, actually issuing an unfavorable unsolicited rating can improve the CRA's reputation because it demonstrates that the agency is willing to resist issuing inflated favorable ratings. These two effects explain why unsolicited ratings can be privately valuable to the CRA.

The paper's key point is subtle: unsolicited ratings are lower than solicited ratings in equilibrium, but they are not necessarily downward biased. The reason is self-selection. High-quality issuers prefer to buy favorable solicited ratings rather than risk an unfavorable unsolicited rating. Low-quality issuers are more likely to end up in the unsolicited-rating pool. Therefore unsolicited ratings look lower on average even without punitive rating deflation.

The welfare effect is ambiguous. If the higher fees induced by the threat of unsolicited ratings dominate the reputational benefit from truthful unfavorable ratings, the CRA may loosen its rating standards and finance more negative-NPV projects. If the fee effect is smaller, unsolicited ratings can make rating standards more stringent by increasing the value of reputation and reducing rating inflation. Thus, the paper does not give a simple pro- or anti-unsolicited-rating policy conclusion. The policy lesson is that unsolicited ratings can discipline rating inflation, but they can also strengthen the CRA's fee-extraction power.

2 Big picture

- **Research question.** Why do rating agencies issue ratings that issuers did not request or pay for, and why are those unsolicited ratings lower on average?
- **Main puzzle.** If unsolicited ratings reveal useful information for free, why would issuers still pay for solicited ratings?
- **Model answer.** The CRA uses unfavorable unsolicited ratings as a credible threat. Issuers buy solicited ratings to avoid being placed in the lower-quality unsolicited pool.
- **Reputation channel.** Issuing an unfavorable unsolicited rating can raise the CRA's reputation by showing that it is not simply selling favorable ratings to fee-paying issuers.
- **Selection channel.** All favorable ratings are solicited in equilibrium, so unsolicited ratings

are associated with lower-quality issuers even without a downward bias.

- **Policy implication.** Allowing unsolicited ratings can raise or lower welfare depending on whether the reputational discipline or fee-extraction effect dominates.

3 Incremental contribution of the paper

- **Theoretical contribution.** The paper provides a dynamic reputation model of the issuer-pays credit rating market that explicitly incorporates unsolicited ratings.
- **Economic explanation.** It explains why unpaid ratings can be profitable for agencies: they support higher fees for paid ratings and can build reputation.
- **Interpretation of empirical fact.** It rationalizes why unsolicited ratings are lower on average without requiring a punitive downward bias.
- **Rating standards.** It derives how unsolicited ratings affect rating inflation and rating stringency.
- **Welfare.** It shows why regulators should not evaluate unsolicited ratings only from the issuer's perspective. The same practice can either improve reputation-based discipline or intensify fee extraction.

4 Model environment

4.1 Agents

The economy contains three types of risk-neutral agents:

- **Issuers.** Firms may have investment projects. Project type is private information to the issuer.
- **Credit rating agency.** The CRA can be ethical or opportunistic. The ethical CRA rates truthfully; the opportunistic CRA chooses a rating policy to maximize expected discounted profit.
- **Investors.** Investors price securities competitively using the rating and the CRA's reputation.

4.2 Project types

There are three issuer states:

- **Good project, $\theta = G$:** succeeds with probability q .
- **Bad project, $\theta = B$:** fails with probability one.
- **No project, $\theta = N$:** the firm has no immediate financing need.

A good project has positive expected value if it receives a favorable rating and can be financed. A bad project is socially wasteful, but it may be financed if it obtains a favorable rating.

4.3 CRA types and reputation

The CRA's type is either ethical or opportunistic. Let μ_t denote investors' belief at the beginning of period t that the CRA is ethical. Reputation matters because it affects the market value of a high rating in the future and therefore the fee the CRA can charge.

The opportunistic CRA may inflate a rating by issuing an H rating to a bad project. This raises current fee income but lowers future reputation if the project fails. The reputation mechanism is imperfect because a failed high-rated project might be interpreted either as bad luck for a good project or as an inflated rating for a bad project.

4.4 Two rating systems

The paper compares two institutional regimes:

1. **Solicited-only system.** The CRA can issue ratings only to firms that request and pay for them.
2. **System with unsolicited ratings.** If a firm rejects a rating offer or does not request a rating, the CRA may issue an unsolicited rating when it has an informative signal.

Solicited ratings use uppercase notation, H and L . Unsolicited ratings use lowercase notation, h and ℓ , reflecting the fact that investors know whether a rating was solicited.

5 Solicited-only rating system

In the solicited-only benchmark, all firms with projects request a rating. The CRA's main choice is whether to offer favorable H ratings only to good firms or also to some bad firms. Let p_t denote the probability that a type B firm receives an H rating from an opportunistic CRA in period t .

The ethical CRA always reports truthfully. Therefore, when investors observe an H rating, the posterior probability that the CRA is ethical and the probability that the project is good both depend on p_t . A larger p_t makes the H rating less informative, lowers the reputation of the CRA after an H rating, and increases the likelihood that high-rated bad projects fail.

5.1 Why solicited-only ratings create inflation

The opportunistic CRA has a direct fee motive to offer H ratings to bad firms. If a bad firm receives an L rating, it will not pay for financing because the rating reveals the bad project. If the CRA instead offers an H rating, the bad firm can raise capital and the CRA can collect a fee. The cost is a possible reputational loss after the bad project fails.

5.2 Equilibrium result

Proposition 1 states that in the solicited-only system there is a unique equilibrium value $p_1 \in (0, 1)$ such that the opportunistic CRA inflates some first-period ratings for bad firms. The CRA offers H ratings to all good firms and to bad firms with positive probability p_1 . Firms invest only if they

obtain an H rating. In the final period, reputation no longer has future value, so the opportunistic CRA gives H ratings to all firms seeking financing.

6 Rating system with unsolicited ratings

The system with unsolicited ratings adds an outside option for the CRA and a new threat for issuers. If a firm does not request a solicited rating, or rejects the offered rating, the CRA can release an unsolicited rating if it has an informative signal. A bad issuer then risks receiving an unfavorable ℓ rating.

6.1 Why an unfavorable unsolicited rating is a credible threat

The threat is credible because issuing an unfavorable unsolicited rating can improve the CRA's reputation. A CRA that issues an ℓ rating shows that it is willing to release bad news rather than sell only favorable ratings. This makes investors update more favorably about the CRA's type.

The threat remains latent in equilibrium. High-quality firms prefer to buy solicited ratings, and bad firms sometimes request ratings to avoid an unsolicited ℓ rating. Therefore the CRA may not need to punish many firms directly; the possibility of punishment supports higher paid-rating fees.

6.2 Equilibrium result

Proposition 2 characterizes an equilibrium in which type G firms always request ratings, while type B firms mix between requesting a rating and not requesting one. The opportunistic CRA offers H ratings to good firms and sometimes to bad firms. If a bad firm refuses a solicited rating, it receives an unsolicited ℓ rating with positive probability when the CRA has an informative signal.

6.3 Selection rather than punishment bias

Corollary 1 is the paper's clean interpretation result. In equilibrium, unsolicited ratings are issued only for type B firms. Therefore, unsolicited ratings are associated with lower firm valuations than solicited ratings. However, this association does not imply downward bias. It arises because favorable ratings are bought as solicited ratings, while firms that end up with unsolicited ratings are selected from the worse issuer pool.

7 Fees, rating inflation, and welfare

7.1 Fee effect

Proposition 3 states that, for a given CRA reputation μ_1 , the fee for a solicited H rating is higher when unsolicited ratings are allowed. The CRA's ability to issue an unfavorable unsolicited rating lowers the issuer's outside option of remaining unrated, making a favorable solicited H rating more valuable.

7.2 Rating-inflation effect

Proposition 4 compares the extent of rating inflation across the two systems. The result is ambiguous. If the CRA captures a large fraction γ of the surplus value created by a rating, then rating inflation can be greater in the solicited-only system. For low values of γ , inflation can be greater in the system with unsolicited ratings.

The intuition is a trade-off. Unsolicited ratings increase the CRA's ability to charge fees, which can make inflated favorable ratings more attractive. But they also make unfavorable ratings reputationally useful, which can discipline inflation. Which effect dominates depends on parameters.

7.3 Welfare effect

Proposition 5 states that if γ is sufficiently large, allowing unsolicited ratings improves social welfare; for low values of γ , it can reduce social welfare. The welfare criterion is the expected NPV of all projects undertaken. If unsolicited ratings reduce rating inflation, they prevent more negative-NPV bad projects from being financed. If they mainly strengthen the CRA's fee extraction and loosen rating standards, welfare can fall.

7.4 Comparative statics

Proposition 6 shows how rating inflation varies with project payoffs, success probabilities, reputation, and CRA bargaining power. A higher payoff R of successful projects increases the value of an H rating and tends to increase the probability that bad firms receive H ratings. A higher CRA reputation can reduce inflation when it sharpens investors' inference from rating outcomes.

8 Notes-created summary tables

8.1 Two systems

Dimension	Solicited-only system	System with unsolicited ratings
Who can be rated	Only issuers that request and pay for ratings.	Issuers can receive ratings even without requesting them, if the CRA observes a signal.
Issuer outside option	Remain unrated.	Risk an unfavorable unsolicited rating.
CRA fee power	Fee comes from value added by a favorable solicited rating.	Fee is higher because the threat of an unfavorable unsolicited rating lowers the outside option.
Reputation channel	Inflated H ratings hurt future reputation after failure.	Unfavorable unsolicited ratings can improve reputation by showing resistance to inflation.
Average unsolicited rating	Not applicable.	Lower than solicited ratings because of self-selection, not necessarily because of downward bias.
Welfare effect	Baseline.	Ambiguous: may improve discipline or increase fee-driven rating inflation.

8.2 Key objects in the model

Object	Meaning
μ_t	Investors' reputation belief at the start of period t : probability that the CRA is ethical.
$\theta \in \{G, B, N\}$	Issuer state: good project, bad project, or no project.
p_t	Probability that an opportunistic CRA gives a solicited H rating to a bad project in the solicited-only system.
$\hat{p}_t$	Analogous probability in the system with unsolicited ratings.
λ_t	Probability that a bad firm requests a rating in the unsolicited-rating system.
δ	Probability that the CRA observes an informative signal when a firm does not request a rating.
γ	Fraction of the surplus value from a rating captured by the CRA as a fee.
R and I	Project payoff if successful and required investment cost.

8.3 Main results

Result	Economic content
Proposition 1	In the solicited-only system, the opportunistic CRA inflates some ratings for bad firms in the first period and all ratings in the last period.
Proposition 2	With unsolicited ratings, good firms request ratings; bad firms may mix; an unfavorable unsolicited ℓ rating can be issued when a bad firm refuses to buy a rating.
Corollary 1	Unsolicited ratings are lower on average because they are issued only to bad-type firms in equilibrium; they are not downward biased.
Proposition 3	Allowing unsolicited ratings increases the fee charged for favorable solicited ratings.
Proposition 4	The effect of unsolicited ratings on rating inflation is ambiguous and depends on the CRA's surplus share γ .
Proposition 5	The welfare effect is ambiguous: unsolicited ratings can raise or lower social welfare.
Proposition 6	Rating inflation responds to project payoff, success probability, reputation, and CRA bargaining power.

8.4 Mechanism map

Mechanism	Effect on CRA behavior	Effect on issuers/investors
Issuer-pays fee motive	Encourages favorable ratings, including inflated ratings for bad firms.	Bad issuers want H ratings because they enable financing.
Reputation concern	Disciplines inflation because failures of high-rated firms damage future rating credibility.	Investors use rating outcomes to update about CRA type.
Unsolicited-rating threat	Raises the fee for favorable solicited ratings.	Issuers buy ratings to avoid the lower outside option of an unsolicited ℓ rating.
Unfavorable-rating signal	Makes issuing an ℓ rating reputationally valuable.	Investors infer that the CRA may be ethical when it releases bad news.
Self-selection	All favorable ratings become solicited.	Unsolicited ratings are lower on average without being punitive.

9 Figures and formal results

9.1 Title, abstract, and motivating puzzle

Read:

- The abstract states the three model ingredients: misreporting, unsolicited ratings, and reputational concerns.
- The introduction explains the empirical puzzle: unsolicited ratings are lower on average, but this need not mean that they are punitive.

Economic message: Unsolicited ratings are both a reputation-building device and a threat that supports solicited-rating fees.

The Economics of Solicited and Unsolicited Credit Ratings

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This paper develops a dynamic rational expectations model of the credit rating process incorporating three critical elements of this industry: (1) the rating agencies' ability to misreport the issuer's credit quality, (2) their ability to issue unsolicited ratings, and (3) the reputational concerns. We analyze the incentives of credit rating agencies to issue unsolicited credit ratings and the effects of this practice on the agencies' rating strategies. We find that issuance of unfavorable unsolicited credit ratings enables rating agencies to extract higher fees from issuers by credibly threatening to punish those that refuse to acquire rating. Also, issuing unfavorable unsolicited ratings increases the rating agencies' reputation by demonstrating to investors that they resist the temptation to issue inflated ratings. In equilibrium, unsolicited credit ratings are lower than solicited ratings, because all favorable ratings are solicited; however, they do not have a downward bias. We show that, under certain conditions, a credit rating system that incorporates unsolicited ratings leads to more stringent rating standards. (JEL C73, D82, D83, G24)

The findings of these papers provide conflicting evidence. On the one hand, using S&P bond ratings on the international market, [Poon \(2003\)](#) reports that issuers who chose not to obtain rating services from S&P have weaker financial profiles; this is consistent with the "self-selection hypothesis." [Garratt \(2004\)](#) finds no significant difference between the performance of issuers with solicited and unsolicited ratings. This result leads her to reject the "punishment hypothesis" in favor of the "self-selection hypothesis." On the other hand, [Bannier, Behr, and Güttler \(2010\)](#) cannot reject the "punishment hypothesis" for their sample.

Our paper suggests an alternative explanation for these findings. We show that although unsolicited ratings are lower, they are not the result of rating deflation. Rather, they reflect the lower quality of issuers, as suggested by the self-selection hypothesis. This does not mean, however, that rating agencies cannot use unfavorable unsolicited ratings as a threat to pressure issuers to pay higher fees for more favorable ratings. We show that the rating agency's ability to issue unfavorable unsolicited ratings to high-quality firms can act as a credible punishment even though it may not be carried out in equilibrium and, hence, may not be observed by investors. This happens because, in equilibrium, the rating agency optimally sets the fee that the agency charges for favorable solicited ratings at a level at which issuers prefer to purchase ratings rather than risk obtaining unfavorable unsolicited ratings.

The remainder of this paper is organized as follows. Section 1 introduces the model. Section 2 describes the equilibrium of the model and analyzes the optimal rating policy in a solicited-only rating system. Section 3 solves for the equilibrium strategies in a rating system that incorporates unsolicited ratings. Section 4 compares the rating agency's fees and rating standards under the two rating systems and derives implications for social welfare. Section 5 summarizes our contribution and concludes. All proofs are contained in the Appendix.

1. The Model

(a) Title and abstract

(b) Empirical puzzle and model overview

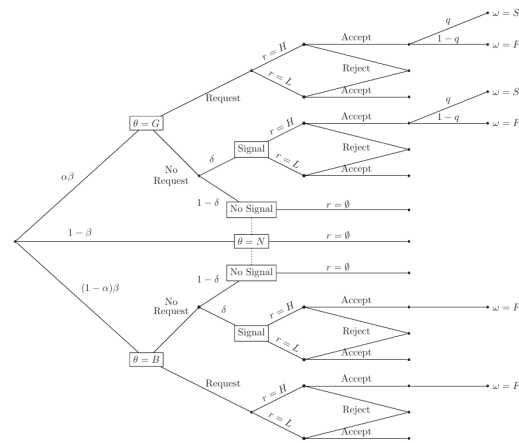
9.2 Decision trees

Read:

- Figure 1 shows the solicited-only system, where rating activity occurs only after a rating request.
- Figure 2 adds the branch where the CRA may observe a signal and issue an unsolicited rating.

Economic message: The only structural difference is the additional post-refusal rating branch, but that branch changes both fee bargaining and reputation incentives.

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(a) Solicited-only decision tree

The Economics of Solicited and Unsolicited Credit Ratings

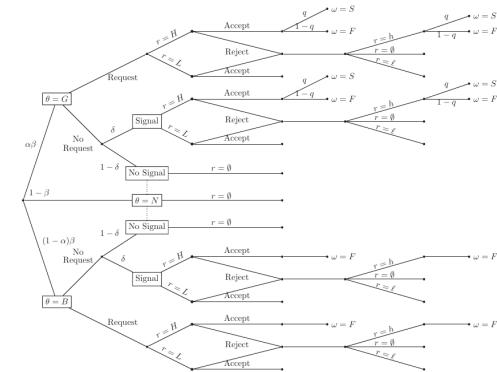


Figure 2
Decision tree in the credit rating system with unsolicited credit ratings

(b) Decision tree with unsolicited ratings

9.3 Reputation and unrated firms in the solicited-only system

Read:

- Equation (1) updates the CRA's reputation after an H rating.
- Equations (2)–(7) show how ratings and lack of ratings affect firm values and investor beliefs.

Economic message: Ratings are valuable only because they convey information about both issuer quality and CRA credibility.

2. The Solicited-Only Credit Rating System

In this section, we solve for the equilibrium in a rating system with solicited ratings only. In this case, the CRA's rating policy in period t is fully characterized by the vector $\{p_t^G, p_t^B\}$, where p_t^B denotes the probability that an H -rating is offered to a firm of type $\theta \in \{G, B\}$.

Absent the option of issuing unsolicited ratings, firms that decline to purchase a rating remain unrated. As we will show below, this applies (in equilibrium) to all firms that are offered an L -rating by the CRA. These firms are better off not acquiring a rating, because an L -rating will reveal that they are of the bad type and, hence, that their value is lower than the value of a firm without a project.²³ Thus, for expositional simplicity, our discussion below focuses only on the case in which the CRA either issues an H -rating or the firm remains unrated. We let ϕ_t denote the fee that the CRA charges for an H -rating.

We also conjecture (and verify) that in equilibrium all firms with an investment project request a rating, thereby revealing their type to the CRA. Further, the CRA offers a favorable H -rating to all type G firms. Thus, whereas "rating inflation" (that is, type B firms receiving an H -rating) will be part of our equilibrium, "rating deflation" (type G firms receiving an L -rating) will not. We therefore simplify our notation by setting $p_t^G = 1$ and by letting p_t denote the probability that an H -rating is offered to a firm known to be of type B .

We begin our analysis by discussing how the CRA's rating policy affects the agency's reputation. Because an ethical CRA always assigns an H -rating (L -rating) to a type G (type B) firm, whereas an opportunistic CRA may follow a different rating policy, the issuance of a credit rating and the subsequent performance of the rated firm is informative about the CRA's type. Investors update their beliefs about the CRA's type twice in each period: first, after the CRA releases a rating, and second, after investors observe the outcome (i.e., success or failure) of the firm's investment project (if an investment has been made).

Let μ_t denote the CRA's reputation in the eyes of investors at the beginning of period $t \in \{1, 2\}$. The CRA's reputation after it issues an H -rating can be derived from Bayes' rule as follows:

$$\mu_t^H \equiv \text{prob}[\tau = e | r_t = H] = \frac{\mu_t \alpha}{\mu_t \alpha + (1 - \mu_t)(\alpha + (1 - \alpha)\bar{p}_t)}, \quad (1)$$

where $\bar{p}_t$ denotes the investors' beliefs about the CRA's rating strategy p_t . Note that issuing an H -rating lowers the CRA's reputation (i.e., $\mu_t^H < \mu_t$) if $\bar{p}_t > 0$. This loss of reputation reflects the fact that, in equilibrium, an H -rating is more likely to be released by an opportunistic CRA than an ethical one. Correspondingly, after observing an H -rating, investors update the probability

that the firm's investment project is of the good type to

$$\alpha_t^H \equiv \text{prob}[\theta = G | r_t = H] = \mu_t^H + (1 - \mu_t^H) \frac{\alpha}{\alpha + (1 - \alpha)\bar{p}_t}. \quad (2)$$

The (gross) market value of an H -rated firm is thus equal to

$$V_t^H = \alpha_t^H q R, \quad (3)$$

which exceeds the amount I invested in the project since $\alpha_t^H \geq \alpha$. It is easy to verify that V_t^H is an increasing function of the CRA's reputation.

Lack of rating activity by the CRA (i.e., the observation of an unrated firm, $r_t = \emptyset$) is also informative about the CRA's type.²⁴ This is because the absence of a rating can mean either that a firm does not have access to an investment project and, hence, does not have any financing needs, or that the CRA offered to issue an L -rating and the firm declined the offer. From Bayes' rule, we have

$$\mu_t^B \equiv \text{prob}[\tau = e | r_t = \emptyset] = \frac{\mu_t(1 - \beta + (1 - \alpha)\beta)}{\mu_t(1 - \beta + (1 - \alpha)\beta) + (1 - \mu_t)(1 - \beta + (1 - \alpha)\beta(1 - \bar{p}_t))}. \quad (4)$$

Equation (4) shows that, if $\bar{p}_t > 0$, lack of a rating increases the CRA's reputation in the eyes of investors (i.e., $\mu_t^B > \mu_t$). This is because it is a signal to investors that the CRA refrained from issuing a potentially inflated rating, a practice that is more likely to happen when the CRA is ethical.

Absence of rating activity also affects the value of unrated firms. The value of an unrated firm to investors is the weighted average of the value of a firm without an investment project (that never requested a rating) and the value of a firm with a project that was offered an L -rating that was then declined. Because the latter category only consists of type B firms that have zero value, the value of an unrated firm in the first period is equal to

$$V_t^U = (1 - \beta_t^U) \bar{V}, \quad (5)$$

where

$$\beta_t^U \equiv \text{prob}[\theta \neq N | r_t = \emptyset] = \mu_t^B \frac{(1 - \alpha)\beta}{1 - \beta + (1 - \alpha)\beta} + (1 - \mu_t^B) \frac{(1 - \alpha)\beta(1 - \bar{p}_t)}{1 - \beta + (1 - \alpha)\beta(1 - \bar{p}_t)}. \quad (6)$$

That is, the probability $1 - \beta_t^U$ represents the investors' updated belief that an unrated firm is of type $\theta = N$. The variable $\bar{V}$ denotes the value of a firm that does not have an investment project in the first period, which is given by

$$\bar{V} = (\alpha + (1 - \alpha)(1 - \mu_t^B)\bar{p}_t) \beta (V_2^H - I - \phi_2), \quad (7)$$

where $V_2^H - I - \phi_2$ is the market value of an H -rated firm in the second

(a) Solicited-only system and reputation after H

(b) Unrated-firm value and belief updating

9.4 Success, failure, and CRA profit

Read:

- Equations (8)–(10) describe reputation after success or failure of an H -rated project and the second-period profit function.
- Equation (11) gives the opportunistic CRA's first-period objective in the solicited-only system.

Economic message: The CRA trades off current rating fees against reputation-dependent future fees.

to be of the opportunistic type (with probability $1 - \mu_1^\theta$).

If an investment is made, which in equilibrium happens only if the firm obtains an H -rating, the project payoff is realized at the end of the period and becomes known to investors. After observing the outcome of the investment project, investors update the CRA's reputation once more. Because firms with good projects are successful with probability q , whereas firms with bad projects always fail, the CRA's updated reputation depends on whether or not the investment project succeeds. If the project succeeds, the firm is revealed as being of type G and the CRA's reputation becomes

$$\mu_t^{H,S} \equiv \text{prob} [\tau = e | r_t = H, \omega_t = S] = \frac{\mu_t \alpha q}{\mu_t \alpha q + (1 - \mu_t) \alpha q} = \mu_t. \quad (8)$$

Project success increases the CRA's reputation (i.e., $\mu_t^{H,S} > \mu_t^H$), because opportunistic CRAs may issue H -ratings with positive probability to bad firms that have a lower success probability. In addition, because in our simplified model only good projects succeed and (in equilibrium) all firms with good projects obtain an H -rating, the observation of a successful H -rated project restores the reputation of the CRA to its original level, that is, $\mu_t^{H,S} = \mu_t$. If the project fails, the CRA's updated reputation is

$$\mu_t^{H,F} \equiv \text{prob} [\tau = e | r_t = H, \omega_t = F] = \frac{\mu_t \alpha (1 - q)}{\mu_t \alpha (1 - q) + (1 - \mu_t) (\alpha (1 - q) + (1 - \alpha) \bar{p}_t)}. \quad (9)$$

Project failure has an adverse effect on the CRA's reputation (i.e., $\mu_t^{H,F} < \mu_t^H$), because an ethical CRA never issues an H -rating for a firm with a bad project. Note that, when updating the CRA's reputation, investors take into account that the failure of an H -rated firm may be the result of "bad luck" (i.e., a good firm failing), rather than of "bad ratings" (i.e., inflated ratings for bad firms). This means that project failure, although negatively affecting the CRA's reputation, does not fully reveal the CRA's type to investors as long as the success probability of good firms, q , is strictly less than one.

We now derive the objective function of the opportunistic CRA. Proceeding backward, in the second (and final) period, the CRA only cares about the profit that it generates from issuing a solicited rating. Thus, the CRA's objective function is

In the first period, the opportunistic CRA chooses a rating policy to maximize the sum of the expected profit obtained in periods 1 and 2:

$$\begin{aligned} \pi_1 = & \alpha \beta \left(\phi_1 + q \pi_2(\mu_1^{H,S}) + (1 - q) \pi_2(\mu_1^{H,F}) \right) \\ & + (1 - \alpha) \beta \left(p_1 \left(\phi_1 + \pi_2(\mu_1^{H,F}) \right) + (1 - p_1) \pi_2(\mu_1^\theta) \right) + (1 - \beta) \pi_2(\mu_1^\theta). \end{aligned} \quad (11)$$

The three components reflect the three cases in which the firm has a good project ($\theta = G$), a bad project ($\theta = B$), or no project ($\theta = N$). If the firm has a good project, the CRA always offers to issue an H -rating and thus earns a fee

(a) Reputation after success/failure and period-2 profit

(b) First-period opportunistic CRA profit

9.5 Proposition 1

Read:

- Proposition 1 gives the equilibrium rating strategy in the solicited-only system.
- The continuation clarifies that the solicited H -rating fee is tied to the surplus value generated by the rating.

Economic message: With only solicited ratings, the CRA sells some inflated favorable ratings because bad issuers value financing and reputation costs are delayed.

Proposition 1. In the solicited-only credit rating system, there exists a unique $p_1 \in (0, 1]$ such that the following strategies are an equilibrium:

- (1) All firms with an investment project request a rating. Firms always acquire an H -rating if they are offered one; they never acquire an L -rating rating. Firms raise funds and invest in the project if and only if they obtain an H -rating.
- (2) In period 1, the opportunistic CRA offers an H -rating to type G firms with probability one and to type B firms with probability $p_1 > 0$; it offers an L -rating to type B firms with probability $1 - p_1$. The fee charged for a

solicited H -rating is $\phi_1 = \gamma(V_1^H - I - V_1^\theta)$. In period 2, the opportunistic CRA offers an H -rating to all firms that seek financing and charges a fee of $\phi_2 = \gamma(V_2^H - I)$.

These strategies are supported by the off-equilibrium beliefs that firms with an L -rating and firms seeking to raise funds without a rating are of type B with

(a) Proposition 1: solicited-only equilibrium

(b) Proposition 1 continuation

9.6 Unsolicited-rating objective and Proposition 2

Read:

- Equations (14)–(16) modify the CRA's objective and the value of being unrated once unsolicited ratings are possible.
- Proposition 2 characterizes the equilibrium with unsolicited ratings.

Economic message: The possibility of an unsolicited ℓ rating changes the outside option of bad firms and makes the bad firm's rating-request decision part of equilibrium.

the agency earns by setting an π -rating to a firm. In the first period, the objective function now takes into account the possibility that the CRA issues an unsolicited ℓ -rating and that (type B) firms may therefore prefer to not request a rating. Thus, the CRA's expected profit in Equation (11) has to be modified as follows:

$$\begin{aligned}\hat{\pi}_1 = & \alpha\beta\left(\hat{\phi}_1 + q\hat{\pi}_2(\hat{\mu}_1^{H,S}) + (1-q)\hat{\pi}_2(\hat{\mu}_1^{H,F})\right) \\ & + \Lambda_1^B\left(\hat{p}_1\left(\hat{\phi}_1 + \hat{\pi}_2(\hat{\mu}_1^{H,F})\right) + (1-\hat{p}_1)\left(\hat{u}_1^B\hat{\pi}_2(\hat{\mu}_1^\ell) + (1-\hat{u}_1^B)\hat{\pi}_2(\hat{\mu}_1^0)\right)\right) \\ & + \Lambda_1^N\hat{\pi}_2(\hat{\mu}_1^0).\end{aligned}\quad (14)$$

The possibility of receiving an unsolicited ℓ -rating also affects the firm's decision of whether to request a rating when its project is of type B . By not requesting a rating, a bad firm may be able to conceal its type from the CRA. In particular, it will be mistaken for a type N firm with probability $1-\delta$, meaning that it will not receive a credit rating in this case. If this happens, its (gross) market value in the first period is equal to

$$\hat{V}_1^0 = (1-\hat{\beta}_1^0)\hat{V} = (1-\hat{\beta}_1^0)\left(\alpha + (1-\alpha)(1-\hat{\mu}_1^0)\hat{p}_2\right)\beta\left(\hat{V}_2^H(\hat{\mu}_1^0) - I - \hat{\phi}_2\right), \quad (15)$$

where $\hat{V}_2^H(\hat{\mu}_1^0) = \hat{\alpha}_2^H(\hat{\mu}_1^0)qR$ and $\hat{\beta}_1^0$ denotes the probability that an unrated firm has an investment project, which is given by

$$\hat{\beta}_1^0 = \hat{\mu}_1^0 \frac{(1-\alpha)\beta(1-\hat{\lambda}_1)(1-\delta)}{\Lambda_1^N} + (1-\mu_1^0) \frac{(1-\alpha)\beta(1-\hat{\lambda}_1)(1-\delta) + \Lambda_1^B(1-\hat{p}_1)(1-\hat{u}_1^B)}{\Lambda_1^N + \Lambda_1^B(1-\hat{p}_1)(1-\hat{u}_1^B)}. \quad (16)$$

Thus, in the first period, the value of an unrated firm exceeds that of an ℓ -rated firm which is zero because such firms are known to be bad. On the other hand

(a) Unsolicited-rating objective and beliefs

Proposition 2. In the credit rating system with unsolicited ratings, if $\beta > \alpha/(1-\alpha)$, there exists a pair $(\hat{p}_1, \hat{\lambda}_1) \in [0, 1]^2$ and a threshold $\bar{\mu} > 0$ such that, for any $\mu_1 < \bar{\mu}$, the following strategies are an equilibrium:

- (1) Type G firms always request a rating; type B firms request a rating with probability $\hat{\lambda}_1$ in period 1, and with probability one in period 2. Firms always acquire an H -rating if they are offered one; they never acquire an L -rating rating. Firms raise funds and invest in the project if and only if they obtain an H -rating.
- (2) In period 1, the opportunistic CRA offers an H -rating to firms known to be of type G with probability one and to firms known to be of type B with probability $\hat{p}_1 \in (0, 1)$; it offers an L -rating to firms known to be of type B with probability $1-\hat{p}_1$. The fee charged for a solicited H -rating is $\hat{\phi}_1 = \gamma(\hat{V}_1^H - I)$. If a firm rejects the offer to acquire a solicited rating, the CRA issues an unsolicited ℓ -rating for the firm with probability one (i.e., $\hat{u}_1^G = \hat{u}_1^B = 1$). In period 2, the opportunistic CRA offers an H -rating to all firms that seek financing and charges a fee of $\hat{\phi}_2 = \gamma(\hat{V}_2^H - I)$.

These strategies are supported by the off-equilibrium beliefs that firms with an L -rating and firms seeking to raise funds without a rating are of type B with probability one, and that a CRA issuing an unsolicited h -rating is of the ethical type with probability μ_1 (i.e., by the passive conjecture). Further, if $I < \alpha^2 qR$, the pair $(\hat{p}_1, \hat{\lambda}_1)$ is unique.

(b) Proposition 2: equilibrium with unsolicited ratings

9.7 Corollary 1 and fee/rating-inflation effects

Read:

- Corollary 1 explains why lower unsolicited ratings arise from equilibrium selection rather than a direct downward bias.
- Propositions 3 and 4 compare fees and rating inflation across the two systems.

Economic message: Unsolicited ratings raise solicited-rating fees, but their effect on rating inflation is parameter-dependent.

investors that the CRA resisted the temptation to issue a possibly inflated H -rating.

It is interesting to note that the beneficial effect of unsolicited ℓ -ratings on the CRA's reputation makes ℓ -ratings a credible threat to firms that refuse to acquire a solicited rating. The threat is credible precisely because these ratings improve the CRA's reputation. This is true for type B firms as well as for type G firms, because neither type of firm can raise the necessary funds to finance its project after receiving an ℓ -rating, which makes any further updating of the CRA's reputation by investors impossible.²⁶ This threat, however, remains "latent" and is not carried out in equilibrium, because all firms are willing to acquire a solicited H -rating (for a fee of $\hat{\phi}_1$) if they are offered one. This means that unsolicited ℓ -ratings are not directly punitive in the sense that they are not issued for good firms, as the following corollary shows.

Corollary 1. In equilibrium, unsolicited ratings are only issued for type B firms. Thus, unsolicited ratings are associated with lower firm valuations,

²⁶ Showing that this argument remains valid in the more general setting in which ℓ -rated firms are still able to obtain financing (and succeed with a (small) positive probability) is straightforward. The reason is that, in equilibrium, investors attribute the success of an ℓ -rated firm to "good luck" rather than to an incorrect rating. This means that the CRA's reputation following the issuance of an ℓ -rating is unaffected by the subsequent observation of a successful project outcome (i.e., $\hat{\mu}_1^L = \hat{\mu}_1^{L,S}$), making the CRA's threat credible even when firms are still able to invest after obtaining an unfavorable unsolicited rating.

We begin with a comparison of the rating fees. The following proposition shows that the ability to issue unsolicited ℓ -ratings allows the opportunistic CRA to charge higher fees for solicited H -ratings.

Proposition 3. For a given reputation μ_1 of the CRA, the fee charged for solicited H -ratings is higher in the rating system that allows for unsolicited ratings than in the solicited-only credit rating system, that is, $\hat{\phi}_1 > \phi_1$.

Proposition 3 provides one of the key insights of this paper. The ability to issue unsolicited ratings is valuable to the CRA because it enables the CRA to charge higher fees and, hence, to extract more surplus from rated firms. The reason is that in the solicited-only credit rating system, firms have the option to avoid a low rating by refusing to be rated by the CRA. In this case, the value of the outside option for a firm is the value of an unrated firm, given by Equation (5). If the CRA has the opportunity to issue unsolicited ratings, firms may no longer have this option. By threatening to issue an unsolicited ℓ -rating for all firms that refuse the CRA's offer to acquire a solicited H -rating, the CRA effectively lowers the value of the firm's outside option to the value of a bad firm, which is zero. This increases the value of an H -rating and, hence, the fee that firms pay for the rating.²⁹

We now turn to a comparison of the extent of rating inflation under the two systems. In our model, the extent of rating inflation can be measured by the probability that a type B firm obtains an H -rating from the opportunistic CRA, which is equal to p_1 in the solicited-only rating system and to $(\hat{\lambda}_1 + (1 - \hat{\lambda}_1)\delta)\hat{p}_1$ in the rating system with unsolicited ratings.

Proposition 4. If the fraction of the surplus value captured by the CRA, γ ,

(a) Corollary 1: unsolicited ratings and selection

(b) Propositions 3 and 4: fees and rating inflation

9.8 Welfare and comparative statics

Read:

- Proposition 5 gives the ambiguous welfare effect of unsolicited ratings.
- Equations (17)–(18) characterize the rating-policy first-order condition in the two systems.

Economic message: Social welfare depends on whether unsolicited ratings mainly strengthen reputation discipline or mainly strengthen fee extraction.

reputation effect associated with releasing unsolicited ℓ -ratings dominates the difference in fees, making rating inflation less attractive to the CRA. Thus, the extent of rating inflation can be less in a system with unsolicited ratings than in a solicited-only system. This result challenges the notion that higher fees due to the issuance of unsolicited ratings compromise the agencies' rating standards.

Assuming that social welfare is utilitarian (i.e., the social welfare function is equally weighted), social welfare in our model equals the expected NPV of all investment projects undertaken by firms. The following result therefore follows immediately from Proposition 4

Proposition 5. If the fraction of the surplus value captured by the CRA, γ , is sufficiently large, the adoption of unsolicited credit ratings leads to an improvement in social welfare. For low values of γ , it can lead to a reduction in social welfare.

Proposition 5 sheds some light on the recent debate on whether or not the adoption of unsolicited ratings should be encouraged, and on how such a change would affect social welfare. Our analysis shows that the answer to these questions depends on the fraction of the firms' surplus extracted by the CRA. When the CRA captures a large part of the surplus (i.e., when γ is high), the issuance of unsolicited ratings leads to less rating inflation and, thus, improves social welfare by preventing firms from investing in negative NPV projects.

We conclude this section by deriving comparative statics results for the opportunistic CRA's rating policy, which can be characterized by the probability with which it offers a favorable H -rating to a firm known to be of type B . The following proposition presents results on how p_1 and $\hat{p}_1$ vary with the model primitives R , q , μ_1 , and γ .

Proposition 6. In both of the credit rating systems, with and without unsolicited ratings, the probability p_1 (respectively, $\hat{p}_1$) that a firm known by the CRA to be of type B is offered an H -rating increases in the payoff R of successful investment projects, increases in the success probability q of type

γ in the rating system with unsolicited ratings.

The opportunistic CRA's rating policy is determined by the first-order condition of the maximization of the CRA's expected profit given by Equation (11) in the solicited-only rating system, and by Equation (14) in the system with unsolicited ratings. In an interior equilibrium, these conditions are given, respectively, by

$$V_1^H - I - V_1^\theta = \beta \left(V_2^H(\mu_1^\theta) - V_2^H(\mu_1^{H,F}) \right), \quad (17)$$

$$\hat{V}_1^H - I = \beta \left(\hat{V}_2^H(\hat{\mu}_1^\ell) - \hat{V}_2^H(\hat{\mu}_1^{H,F}) \right), \quad (18)$$

where $V_2^H(\mu_2)$ (respectively, $\hat{V}_2^H(\mu_2)$) denotes the value of an H -rated firm in period 2, given that the CRA's reputation equals μ_2 . Inspection of Equations (17) and (18) reveals that the equilibrium rating policy of the opportunistic CRA trades off the immediate benefit of earning a fee of $V_1^H - I - V_1^\theta$ (respectively, $\hat{V}_1^H - I$) from issuing an H -rating for a type B firm against a higher future fee due to an improved reputation from not issuing a rating (right-hand side of Equation (17)) or from issuing an unsolicited ℓ -rating (right-hand side of Equation (18)). This trade-off is affected by the model parameters as follows

(a) Propositions 5 and 6: welfare and comparative statics

(b) Equations (17)–(18): rating-policy trade-off

9.9 Conclusion

Read:

- The conclusion restates the two reasons for unsolicited ratings: fee extraction through a credible threat and reputation improvement through unfavorable ratings.
- It also restates the central caution: unsolicited ratings need not be downward biased, but they can still lower welfare if they reduce rating stringency.

Economic message: The same institutional practice can discipline ratings or distort ratings, depending on the strength of the fee and reputation channels.

these ratings are, on average, lower than solicited ratings. We analyze the implications of this practice for credit rating standards, rating fees, and social welfare. Our model incorporates three critical elements of the credit rating industry: (1) the rating agencies' ability to misreport the issuer's credit quality, (2) their ability to issue unsolicited ratings, and (3) their reputational concerns. We focus on a monopolistic rating agency that interacts with a series of potential issuers. In equilibrium, the agency trades off a higher short-term profit from selling inflated ratings to low-quality issuers against a lower long-term profit associated with a reduction in the agency's reputation.

Our analysis shows that the rating agency issues unsolicited ratings for two reasons. First, doing so enables the rating agency to charge higher fees for solicited ratings, because the agency can credibly threaten to punish issuers that refuse to solicit a rating with an unfavorable unsolicited rating. This increases the value of a favorable rating and, hence, the fee that an issuer is willing to pay for it. Second, by issuing a low unsolicited rating, the rating agency can demonstrate to investors that it resists the temptation to issue inflated ratings, thereby improving its reputation.

We demonstrate that, in equilibrium, unsolicited ratings are lower than solicited ratings, because all favorable ratings are solicited. This does not mean, however, that unsolicited ratings have a downward bias. Rather, they reflect the lower quality of firms that do not request a rating.

Comparing credit rating systems with and without unsolicited ratings, we find that although rating agencies benefit from having the option to issue unsolicited ratings, such a system can actually lead to less stringent credit rating standards, thereby reducing social welfare.

Conclusion summary

10 How to remember the paper

- **One-sentence version.** Unsolicited ratings let a CRA credibly threaten issuers and build reputation, so they can raise solicited-rating fees and affect rating inflation.
- **Most important distinction.** Lower average unsolicited ratings do not automatically mean downward bias; they can be lower because good firms self-select into solicited ratings.
- **Key mechanism.** An unfavorable unsolicited rating is credible because it can improve the CRA's reputation.
- **Key trade-off.** More fee power can worsen rating inflation, while more reputation value can reduce it.
- **Policy takeaway.** Whether unsolicited ratings should be encouraged is ambiguous: the sign depends on model parameters, especially the CRA's share of rating-created surplus.

11 Conclusion

11.1 Core conclusion

The paper explains unsolicited ratings as an equilibrium institution in an issuer-pays rating market. The CRA may issue them not because they directly generate fees, but because they make solicited ratings more valuable and can demonstrate the CRA's willingness to release unfavorable information.

11.2 Why the result matters

The paper clarifies an important empirical interpretation problem. Observing that unsolicited ratings are lower than solicited ratings is not sufficient evidence that agencies punish nonpaying issuers with biased ratings. The lower average rating can arise from self-selection: issuers with favorable prospects buy solicited ratings, while lower-quality issuers are more likely to be rated unsolicited.

11.3 Takeaway

Unsolicited ratings are a double-edged instrument. They may discipline rating inflation through reputation, but they may also expand the CRA's fee-extraction power. The welfare effect is therefore not mechanical and must be evaluated through the relative strength of these two channels.